EXPERIMENT - 6

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6.1.1] Incremented Date

A) ALGORITHM

1. Start.
2. Read day.
3. Read month.
4. Read year.
5. If year ≤ 0, print "Invalid Date" and Stop.
6. If month < 1 or month > 12, print "Invalid Date" and Stop.
7. If month is 1, 3, 5, 7, 8, 10, 12 then max_days = 31.
8. Else if month is 4, 6, 9, 11 then max_days = 30.
9. Else if month is 2:

   If (year % 400 == 0) OR (year % 4 == 0 AND year % 100 ≠ 0) max_days = 29 Else

   max_days = 28
10. If day < 1 or day > max_days, print "Invalid Date" and Stop.
11. If day < max_days then day = day + 1. 12. Else day = 1 If month == 12 month = 1 year = year + 1 Else

    month = month + 1
13. Print day-month-year in format DD-MM-YYYY.
14. Stop.

B) PYTHON CODE

```
day = int(input())

month = int(input())

year = int(input())


def is_leap(year):     if (year % 400 == 0) or (year % 4 == 0 and

year % 100 != 0):
```

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```
    return True
return False


def days_in_month(month, year):
  if month in [1, 3, 5, 7, 8, 10, 12]:
    return 31   elif month
in [4, 6, 9, 11]:
    return 30
elif month == 2:
    return 29 if is_leap(year) else 28
else:    return 0


if year <= 0 or month < 1 or month > 12:
  print("Invalid Date") elif day < 1 or day >
days_in_month(month, year):
  print("Invalid Date") else:  day += 1  if
day > days_in_month(month, year):
    day = 1
month += 1    if
month > 12:
month = 1
year += 1
  print(f"{day:02d}-{month:02d}-{year}")
```

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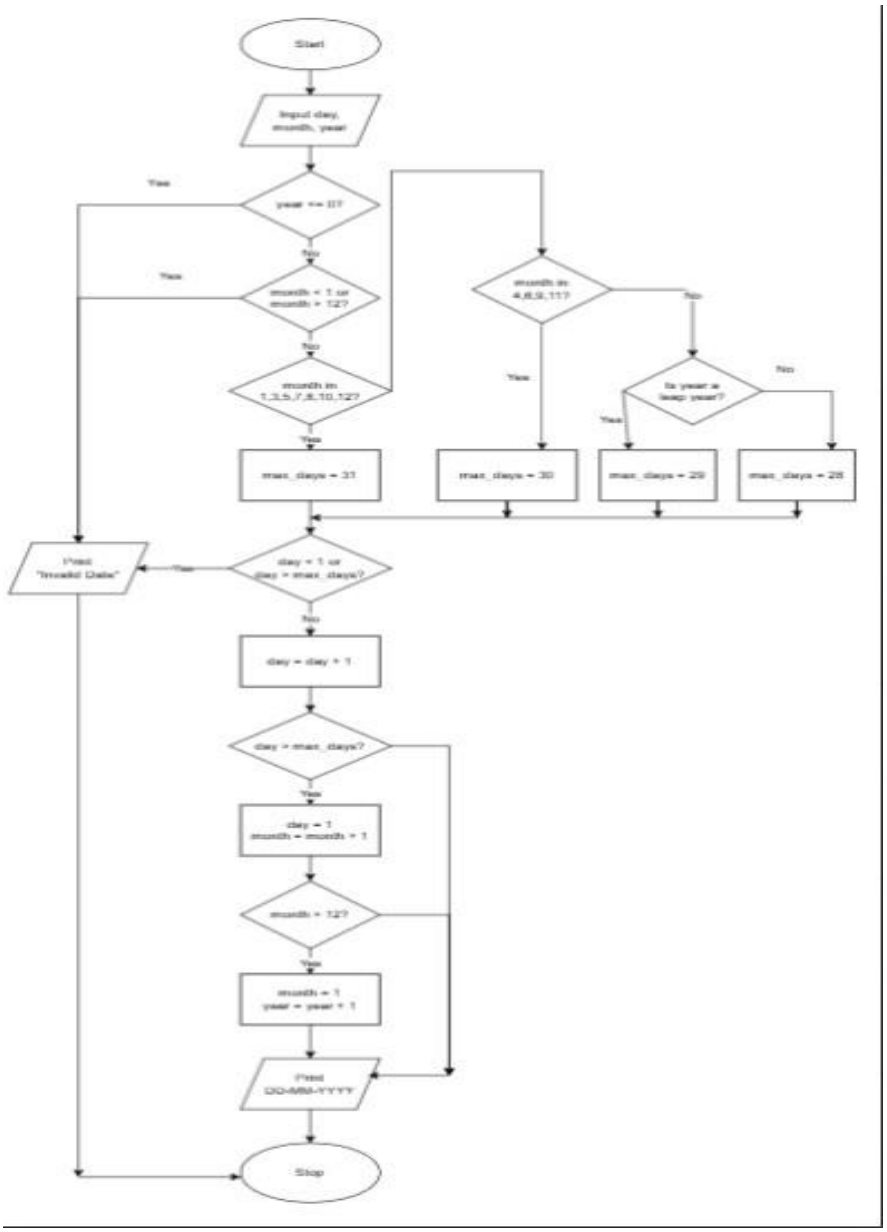
C) FLOWCHART



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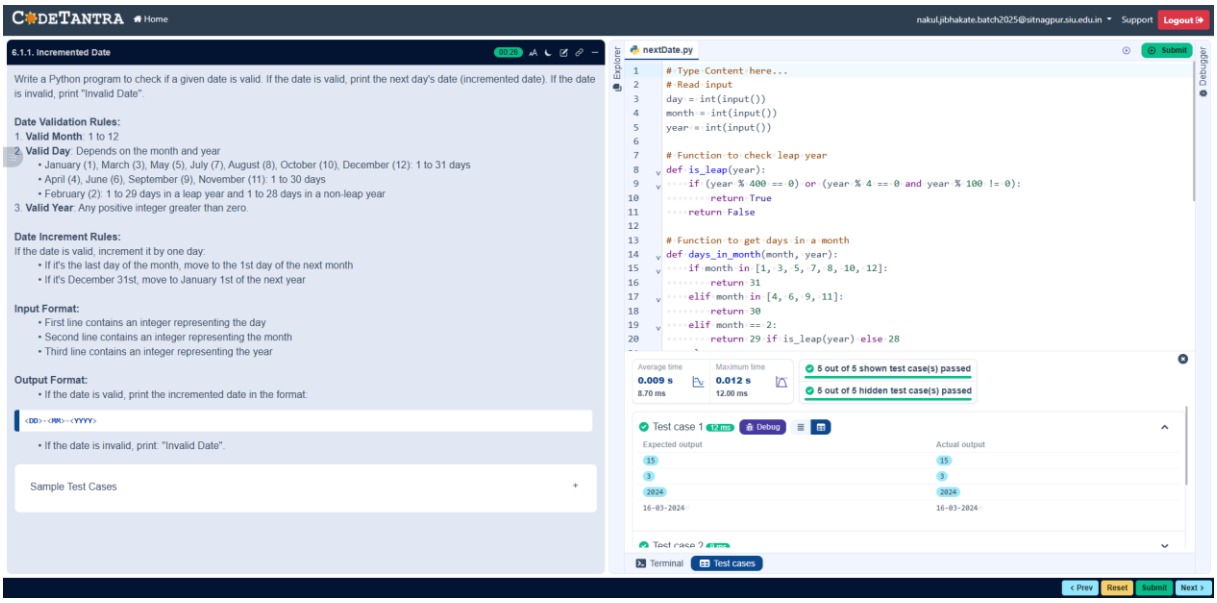
D) EXCECUTION